Assign10 outputs
Q1

```
Booting from Hard Disk...xv6...
cpu0: starting 0
sb: size 1000 nblocks 941 ninodes 200 nlog 30 logstart 2 
init: starting sh
$ memtest

Q1: Physical Memory Allocation Test

Before allocation:

--- Physical Memory Statistics ---
Total allocated pages: 425
Total freed pages: 214
Currently allocated pages: 211
Available physical memory pages: 56790

Allocating 5 pages...

After allocation:

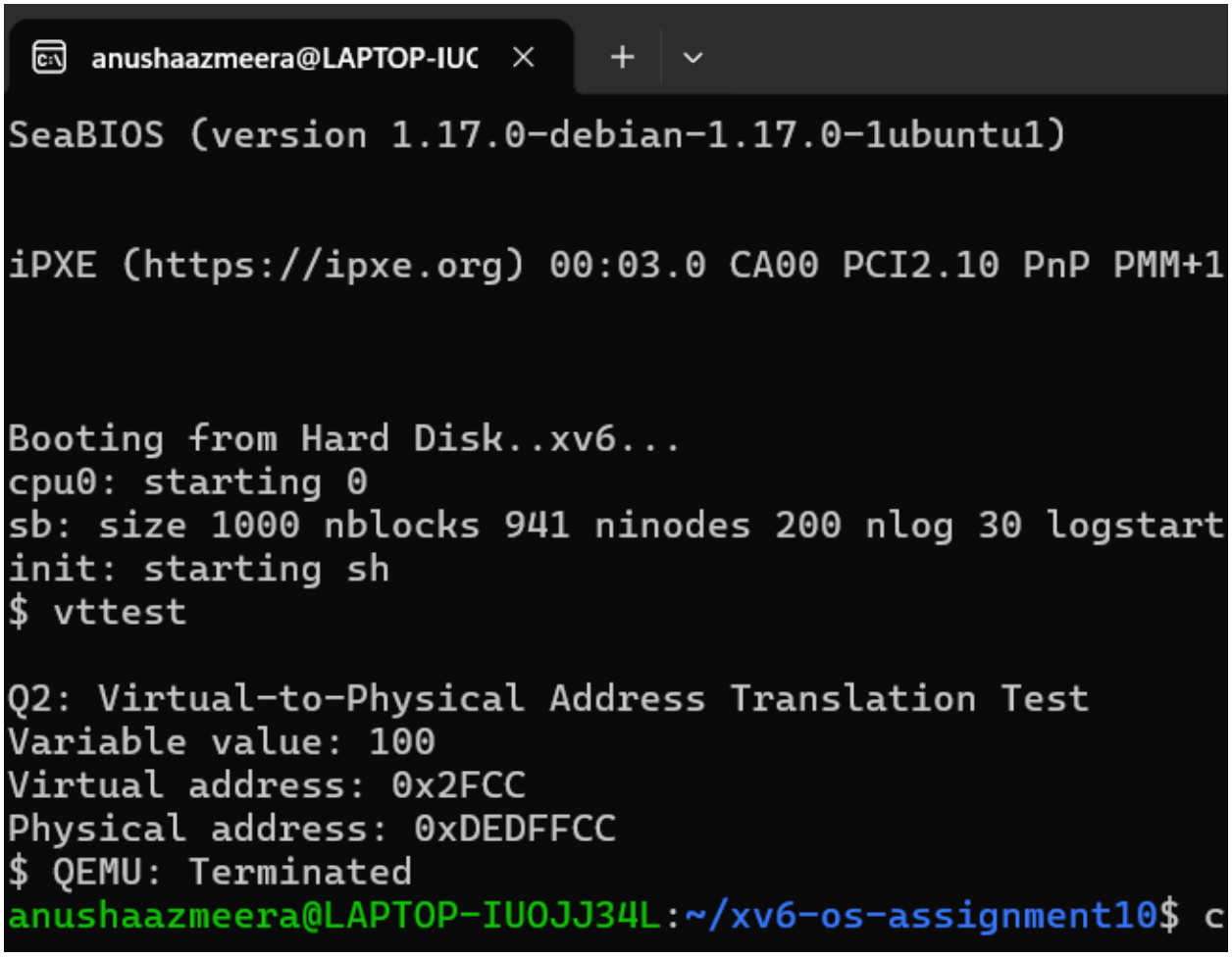
--- Physical Memory Statistics ---
Total allocated pages: 430
Total freed pages: 214
Currently allocated pages: 216
Available physical memory pages: 56785

Releasing 5 pages...

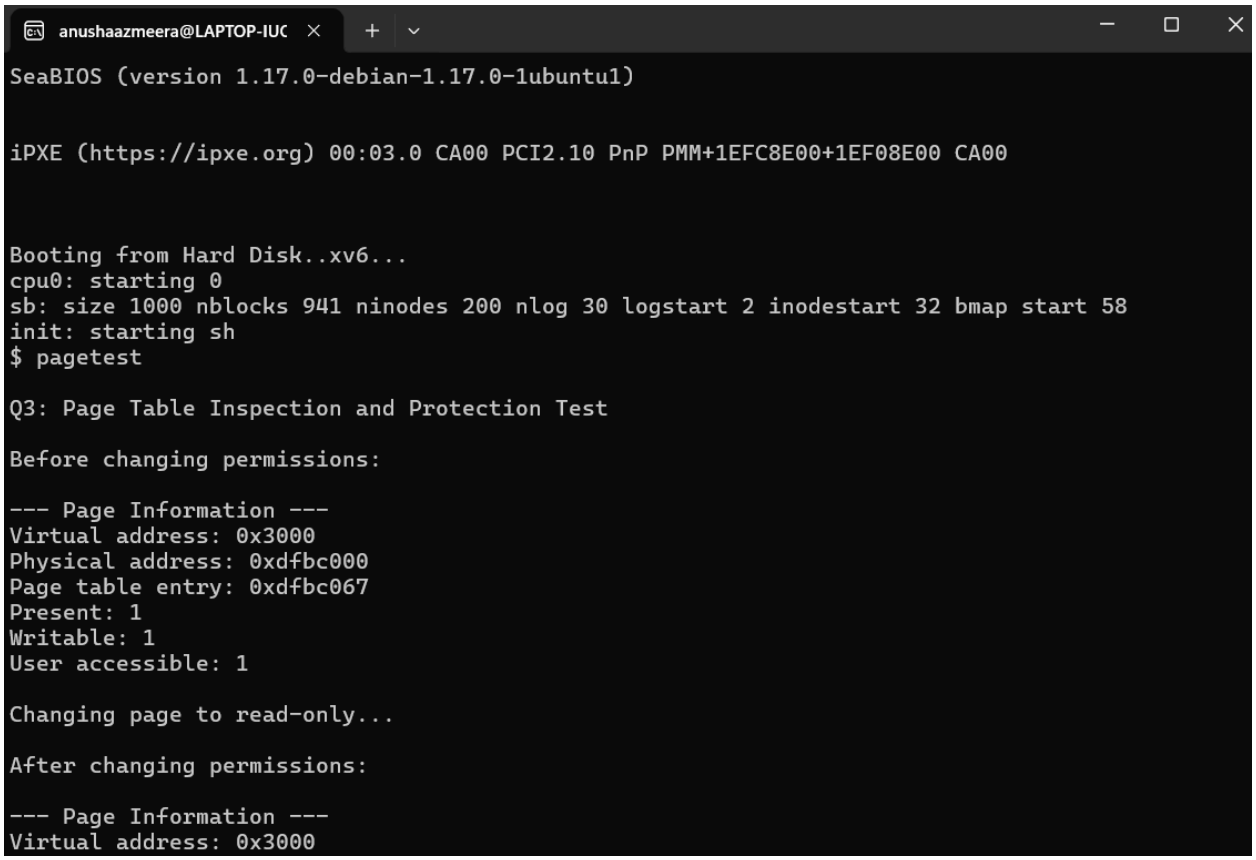
After releasing memory:

--- Physical Memory Statistics ---
Total allocated pages: 430
Total freed pages: 219
Currently allocated pages: 211
Available physical memory pages: 56790
$ 
```

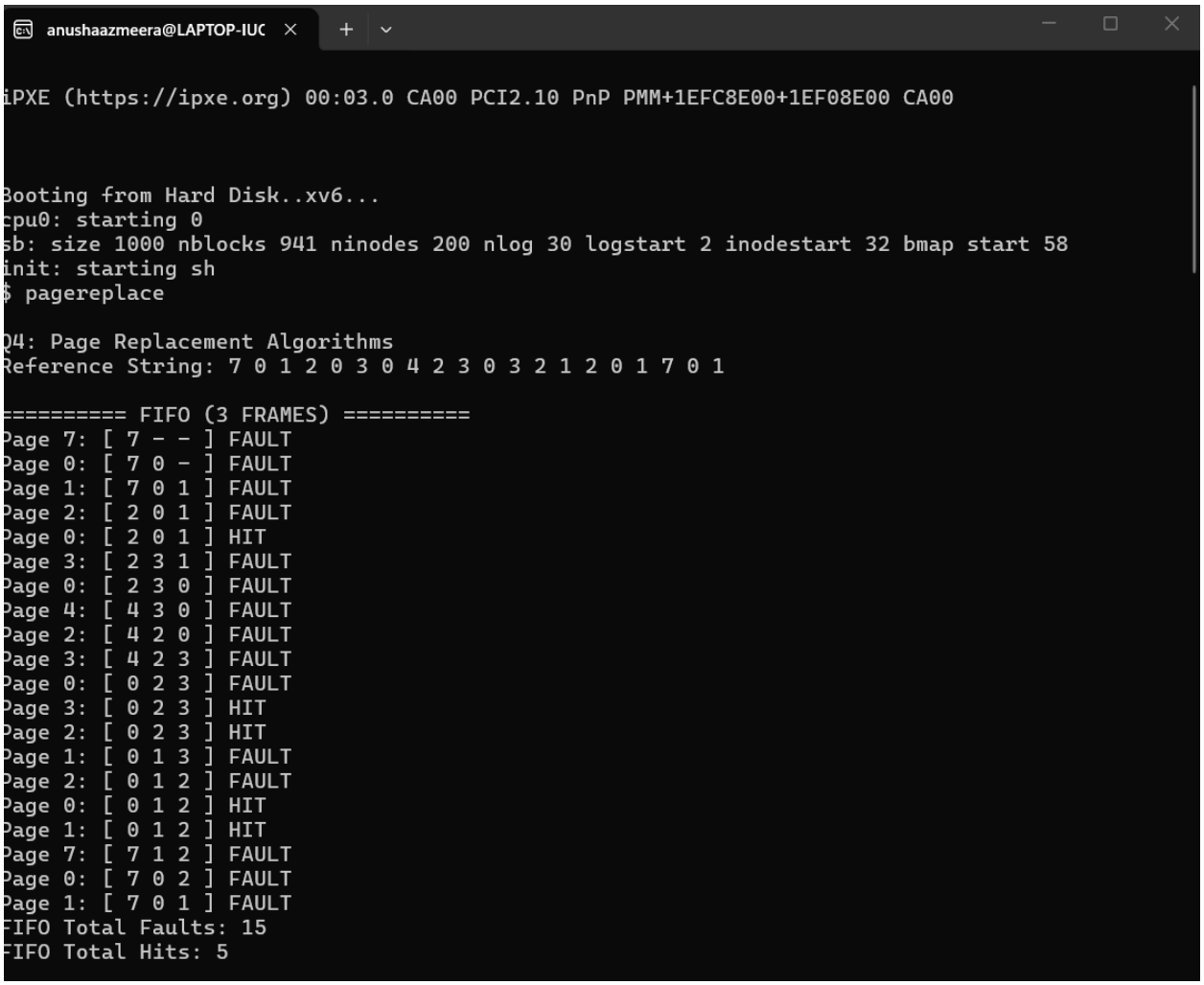
Q2


anushaazmeera@LAPTOP-IUC
SeaBIOS (version 1.17.0-debian-1.17.0-1ubuntu1)
iPXE (https://ipxe.org) 00:03.0 CA00 PCI2.10 PnP PMM+1
Booting from Hard Disk..xv6...
cpu0: starting 0
sb: size 1000 nblocks 941 ninodes 200 nlog 30 logstart
init: starting sh
$ vttest
Q2: Virtual-to-Physical Address Translation Test
Variable value: 100
Virtual address: 0x2FCC
Physical address: 0xDEDFFCC
$ QEMU: Terminated
anushaazmeera@LAPTOP-IUOJJ34L:~/xv6-os-assignment10$ c

Q3


anushaazmeera@LAPTOP-IUC
SeaBIOS (version 1.17.0-debian-1.17.0-1ubuntu1)
iPXE (https://ipxe.org) 00:03.0 CA00 PCI2.10 PnP PMM+1EFC8E00+1EF08E00 CA00
Booting from Hard Disk..xv6...
cpu0: starting 0
sb: size 1000 nblocks 941 ninodes 200 nlog 30 logstart 2 inodestart 32 bmap start 58
init: starting sh
$ pagetest
Q3: Page Table Inspection and Protection Test
Before changing permissions:
--- Page Information ---
Virtual address: 0x3000
Physical address: 0xdfbc000
Page table entry: 0xdfbc067
Present: 1
Writable: 1
User accessible: 1
Changing page to read-only...
After changing permissions:
--- Page Information ---
Virtual address: 0x3000

Q4


anushaazmeera@LAPTOP-IUC
iPXE (https://ipxe.org) 00:03.0 CA00 PCI2.10 PnP PMM+1EFC8E00+1EF08E00 CA00
Booting from Hard Disk..xv6...
cpu0: starting 0
sb: size 1000 nblocks 941 ninodes 200 nlog 30 logstart 2 inodestart 32 bmap start 58
init: starting sh
$ pagereplace
Q4: Page Replacement Algorithms
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1
========== FIFO (3 FRAMES) ==========
Page 7: [7 - -] FAULT
Page 0: [7 0 -] FAULT
Page 1: [7 0 1] FAULT
Page 2: [2 0 1] FAULT
Page 0: [2 0 1] HIT
Page 3: [2 3 1] FAULT
Page 0: [2 3 0] FAULT
Page 4: [4 3 0] FAULT
Page 2: [4 2 0] FAULT
Page 3: [4 2 3] FAULT
Page 0: [0 2 3] FAULT
Page 3: [0 2 3] HIT
Page 2: [0 2 3] HIT
Page 1: [0 1 3] FAULT
Page 2: [0 1 2] FAULT
Page 0: [0 1 2] HIT
Page 1: [0 1 2] HIT
Page 7: [7 1 2] FAULT
Page 0: [7 0 2] FAULT
Page 1: [7 0 1] FAULT
FIFO Total Faults: 15
FIFO Total Hits: 5

```
========== LRU (3 FRAMES) ==========
Page 7: [ 7 - - ] FAULT
Page 0: [ 7 0 - ] FAULT
Page 1: [ 7 0 1 ] FAULT
Page 2: [ 2 0 1 ] FAULT
Page 0: [ 2 0 1 ] HIT
Page 3: [ 2 0 3 ] FAULT
Page 0: [ 2 0 3 ] HIT
Page 4: [ 4 0 3 ] FAULT
Page 2: [ 4 0 2 ] FAULT
Page 3: [ 4 3 2 ] FAULT
Page 0: [ 0 3 2 ] FAULT
Page 3: [ 0 3 2 ] HIT
Page 2: [ 0 3 2 ] HIT
Page 1: [ 1 3 2 ] FAULT
Page 2: [ 1 3 2 ] HIT
Page 0: [ 1 0 2 ] FAULT
Page 1: [ 1 0 2 ] HIT
Page 7: [ 1 0 7 ] FAULT
Page 0: [ 1 0 7 ] HIT
Page 1: [ 1 0 7 ] HIT
LRU Total Faults: 12
LRU Total Hits: 8
```

```
========== OPTIMAL (3 FRAMES) ==========
Page 7: [ 7 - - ] FAULT
Page 0: [ 7 0 - ] FAULT
Page 1: [ 7 0 1 ] FAULT
Page 2: [ 2 0 1 ] FAULT
Page 0: [ 2 0 1 ] HIT
Page 3: [ 2 0 3 ] FAULT
Page 0: [ 2 0 3 ] HIT
Page 4: [ 2 4 3 ] FAULT
Page 2: [ 2 4 3 ] HIT
Page 3: [ 2 4 3 ] HIT
Page 0: [ 2 0 3 ] FAULT
Page 3: [ 2 0 3 ] HIT
Page 2: [ 2 0 3 ] HIT
Page 1: [ 2 0 1 ] FAULT
Page 2: [ 2 0 1 ] HIT
Page 0: [ 2 0 1 ] HIT
Page 1: [ 2 0 1 ] HIT
Page 7: [ 7 0 1 ] FAULT
Page 0: [ 7 0 1 ] HIT
Page 1: [ 7 0 1 ] HIT
Optimal Total Faults: 9
Optimal Total Hits: 11

========== PAGE REPLACEMENT COMPARISON ==========
Frames  Algorithm       Faults  Hits    Hit Ratio       Fault Ratio
3       FIFO            15      5       5/20            15/20
3       LRU             12      8       8/20            12/20
3       Optimal         9       11      11/20           9/20
4       FIFO            10      10      10/20           10/20
4       LRU             8       12      12/20           8/20
4       Optimal         8       12      12/20           8/20
5       FIFO            9       11      11/20           9/20
5       LRU             7       13      13/20           7/20
5       Optimal         7       13      13/20           7/20
$
```